



*In the figments of your imagination, a world appears —
picturesque scenes that flow like a river. And every second, it fades.
The pen is too slow for the palace in your mind.*

Our essays have autocomplete. Shouldn't our imagination?

MaoBi — autocomplete for drawing, and the suite that changes how thought
reaches creation.

Creation is not the bottleneck. Delivery of thought to creation is.

- › AI made creation cheap. The human→AI input channel did not change: type text, describe, wait.
- › We change the input channel itself → **MaoBi** and the related suite.
- › Text got autocomplete years ago. Imagination did not.

THE PROBLEM

Fast but fading, or faithful but slow

Ideas fade faster than hands move. Sketching is the least restrictive way to get an idea out of your head — but every way of capturing ideas takes something.

Pen & paper

Fast and free — but the result is dead ink. Not structured, not editable, not shareable.

3

Design tools

Real structure — but menus, composition, precision, and slow thinking.

Text → image AI

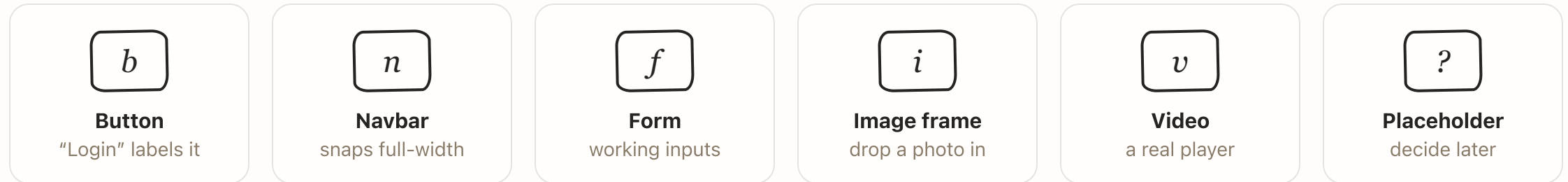
A pretty picture — uneditable, detached from what you drew, no spatial control.

MaoBi

The speed of sketching, the structure of a design tool. Fast and faithful — for the first time.

MaoBi — autocomplete for drawing

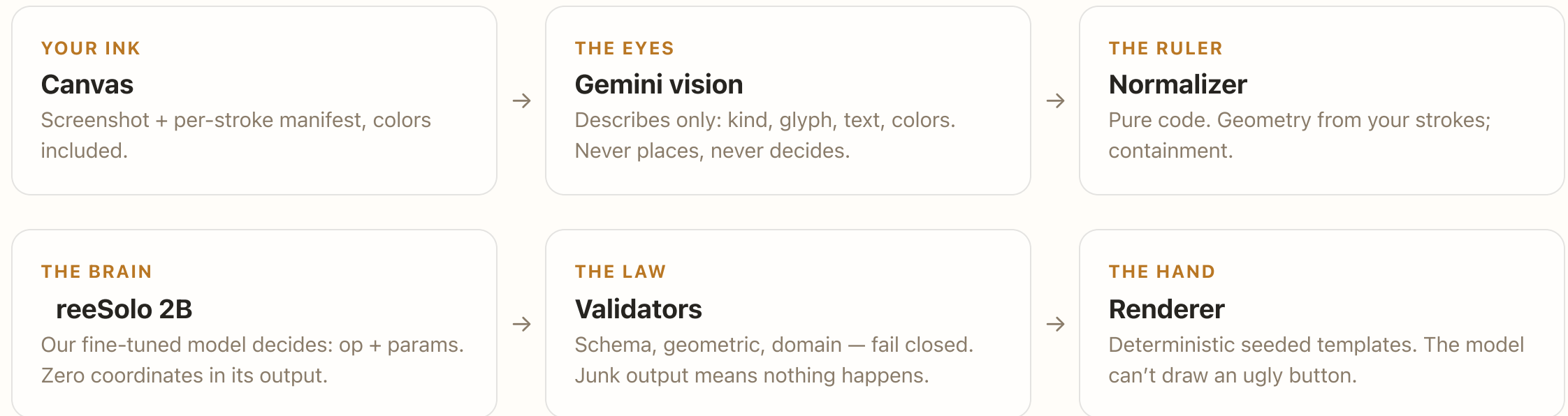
Sketch a page; real editable components bloom in exactly where you drew them. Plain shapes stay plain — a single letter adds function; words and colors add style.



- › The sketch is the spec: one press turns the page into a working, downloadable website or React project.
- › Low confidence → the model abstains; ink stays ink. Failure mode is "nothing happens," never "wrong thing happens."

Two models, strict separation of powers

No model is ever trusted with geometry. Every hand-off is validated — a hallucination becomes nothing, never a broken page.



Zero coordinates, by design. The model never says *where* — geometry derives from your actual ink, so nothing can ever be misplaced.

We trained the brain ourselves — and it beats Gemini

Qwen3.5-2B, LoRA-fine-tuned on FreeSolo against our own synthetic sketch dataset. Untouched held-out test split, our model vs the Gemini Flash baseline.

96.7%

Op accuracy

vs 75.0% Gemini Flash baseline

+33

Detail routing (pts)

word-in-box → label, colors → fill: 90–93.
vs 58.7%

100%

Night-sky-from-rect

vs 25% baseline

0.97

Abstention F1

vs 0.67 — it knows when to stay quiet

<\$0.25

Total training spend

~10 runs · 640 examples × 4 epochs · LoRA
SFT

0

Broken pages possible

0–1.7% hallucination, all caught by fail-closed
validators

We are the user

- › **Founder-workflow moat:** the team has lived this loop countless times and knows what builders want.
 - Building with AI and refining the process since summer 2025 — the first wave of AI coding.
 - Prior researchers, hackathon organizers, hackers, builders (5× hackathons won).
 - Natively developed AI-integrated development workflows at internships.
- › **Data flywheel:** every accept / reject / relabel is a gold training label.
- › **Own model, not API glue:** our 2B beats the frontier baseline on our task, trained for cents.

Waterloo first

- › **Waterloo:** hackathons + a dense population of student builders.
- › **Hackathon distribution:** demo-first product, zero learning curve, spreads at build events.



The inp

- › **Domain-a**
vocabulary
- › **Hybrid input**
 - Magic wand
 - Text boxes
 - Text desc
- › **Canvas as p**
- › **Wires and pages as code abstraction** — code abstraction in general.
- › Edge AI. Tablet/whiteboard SDK. Design-tool export.



*Start the drawing.
MaoBi finishes the thought.*

Paper is fast but dead. Design tools are structured but slow. MaoBi is both: paper with magic in it.